

Cloudflare Radar 2024 Year in Review

A data-driven analysis of global Internet traffic, adoption trends, connectivity disruptions, and security threats

Observed through Cloudflare's global network | 2024

17% Global Traffic Growth and Starlink's 3.3x Surge

Global Internet Traffic Growth

17%

Year-over-year increase in 2024, continuing the upward trend driven by increasing reliance on the Internet for communication, commerce, entertainment, and transportation.

SpaceX Starlink Traffic Surge

3.3x

Dramatic growth during 2024, continuing to bring satellite Internet connectivity to previously underserved locations. Trend scaled relative to peak observed request volume.

Normalized Starlink traffic worldwide



Google Leads Top 10 Services as Generative AI Rises

Top 10 Most Popular Internet Services

Rank	Service
#1	Google
#2	Facebook
#3	Apple
#4	TikTok
#5	AWS
#6	Microsoft
#7	Instagram
#8	YouTube
#9	Amazon
#10	WhatsApp

Generative AI on the Rise

Generative AI continued its rapid rise throughout 2024, while the Metaverse receded into the background. Google maintained its #1 position for the second consecutive year. Rankings based on aggregate data from Cloudflare's 1.1.1.1 DNS resolver.

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Post-Quantum Encryption Jumps from 2% to 13% of TLS 1.3

Post-Quantum Encryption

2% → 13%

Of TLS 1.3 traffic after Chrome 124 enabled post-quantum key agreement by default on April 17, 2024.

HTTP/3 Adoption

21%

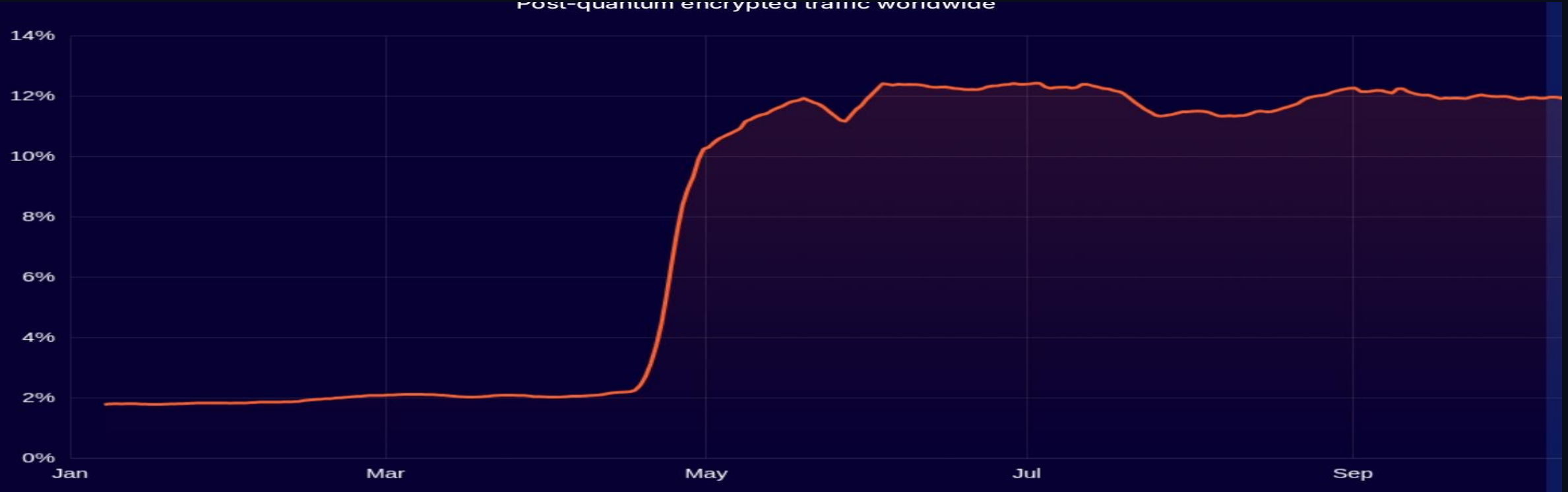
Of traffic uses HTTP/3, running on QUIC for faster connections, packet-loss mitigation, and encryption by default.

Top API Client Language

Go

~12% of automated API requests. Followed by Node.js, Python (9.6%), Java (7.4%), and .NET (3.6%).

Post-quantum encrypted traffic worldwide



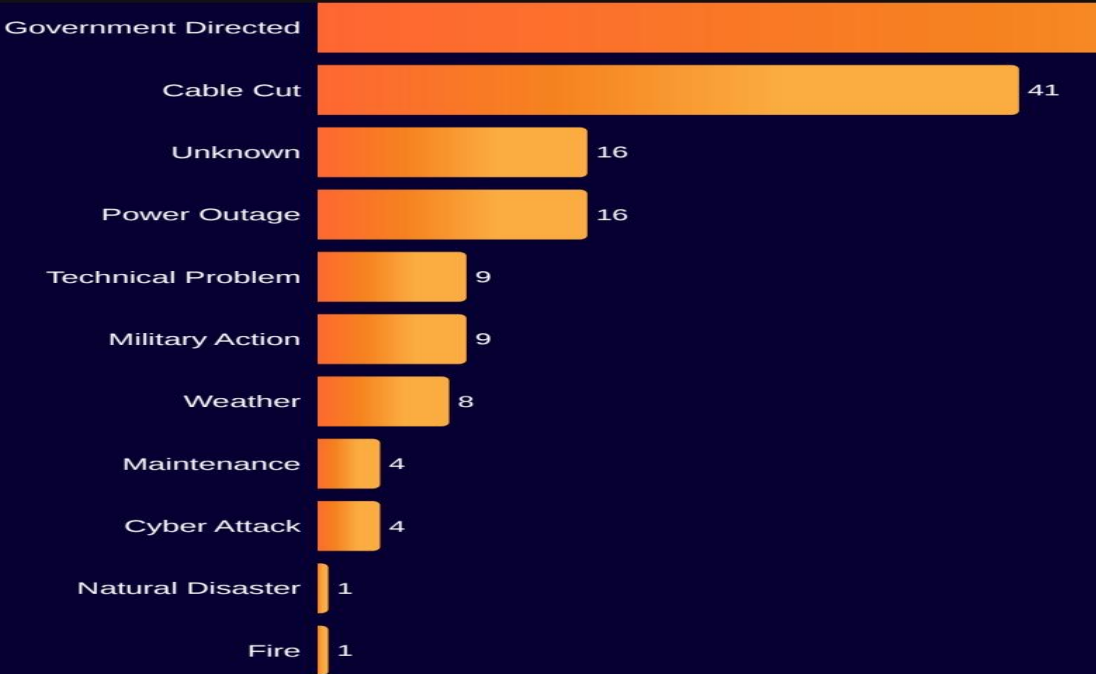
225 Major Disruptions: Government Shutdowns Lead Outage Causes

Causes of Major Internet Disruptions (2024)

Cause	Count
Government Directed	116+
Cable Cut	41
Unknown	16
Power Outage	16
Technical Problem	9
Military Action	9
Weather	8
Maintenance	4
Cyber Attack	4
Natural Disaster	1
Fire	1

Key Insight

Government-directed shutdowns accounted for over half (116+) of the 225 major Internet disruptions observed globally in 2024. Cable cuts (41) and power outages (16) were the next most frequent causes, underscoring the fragility of physical and political infrastructure.



6.5% of Traffic Mitigated: Persistent Threats and Bot Concentration

Traffic Mitigated

6.5%

Of all traffic processed by Cloudflare was mitigated, indicating the scale of ongoing threats.

Most Attacked Sector

**Gambling &
Games**

The most attacked industry category across Cloudflare's protected customer base.

Bot Traffic Concentration

68%

Of global bot traffic originated from just the top 10 countries.

Log4j Persistence (CVE-2021-44228)

First disclosed in December 2021, Log4j continues to be actively targeted by attackers more than two years later — underscoring the persistent nature of widely known vulnerabilities in production systems.

Email Security Threats

4.3% of emails processed were malicious

Among malicious emails, 43% contained a deceptive link. The .bar TLD originated the largest share of malicious and spam email observed during the year.

Routing Security: RPKI (Resource Public Key Infrastructure) valid routes increased by 6.4% — an important step toward improved Internet routing security.

Five Takeaways Shaping the 2025 Internet Outlook

1. Traffic Growth Trajectory

Global Internet traffic grew 17% year-over-year, with Starlink traffic surging 3.3x as satellite connectivity expands to underserved regions.

2. Post-Quantum Encryption Momentum

Adoption leapt from ~2% to 13% of TLS 1.3 traffic following Chrome 124's default enablement in April 2024. Firefox and Safari are now rolling out support.

3. Connectivity Fragility from Government Actions

Government-directed shutdowns remained the leading cause of the 225 major Internet disruptions observed, followed by cable cuts (41) and power outages (16).

4. Persistent Security Threats

6.5% of all traffic was mitigated, Gambling/Games was the most targeted sector, and Log4j exploitation persists years after disclosure.

5. Evolving Developer Ecosystem

Go surpassed other languages to become the most popular API client